

# Bangabandhu Sheikh Mujibur Rahman University, Kishoreganj

1<sup>st</sup> Year 2<sup>nd</sup> Semester B. Sc. (Engg.) Final Examination, Session: 2021-22

Department of Computer Science and Engineering

EEE 1206: Electrical Circuits Lab (1.5 Credits)

Time: 3 Hours

Marks: 40

## Answer only one question

1. Apply Ohm's law and calculate the unknown resistance of the resistors used in the following circuit. Iterate the process for two test cases. In the first test case, consider,  $V_s=10V$ , and in the second test case, consider,  $V_s=15V$ .

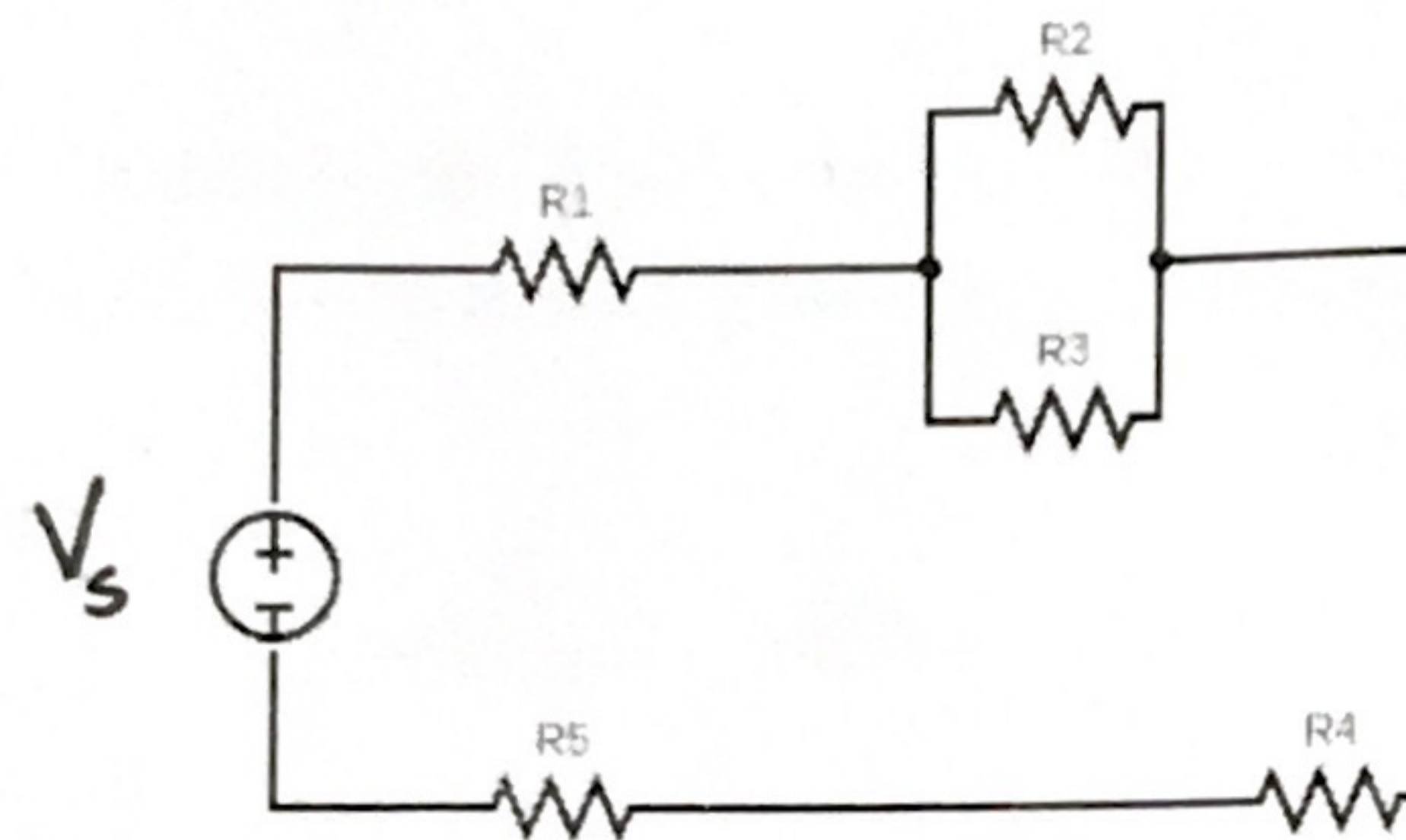


Fig.1. Circuit for Verification of the Ohm's Law

2. Verify Kirchhoff's Voltage law and Voltage Divider Rule for the following circuit. Iterate the process for two test cases. In the first test case, consider,  $V_s=10V$ , and in the second test case, consider,  $V_s=15V$ .

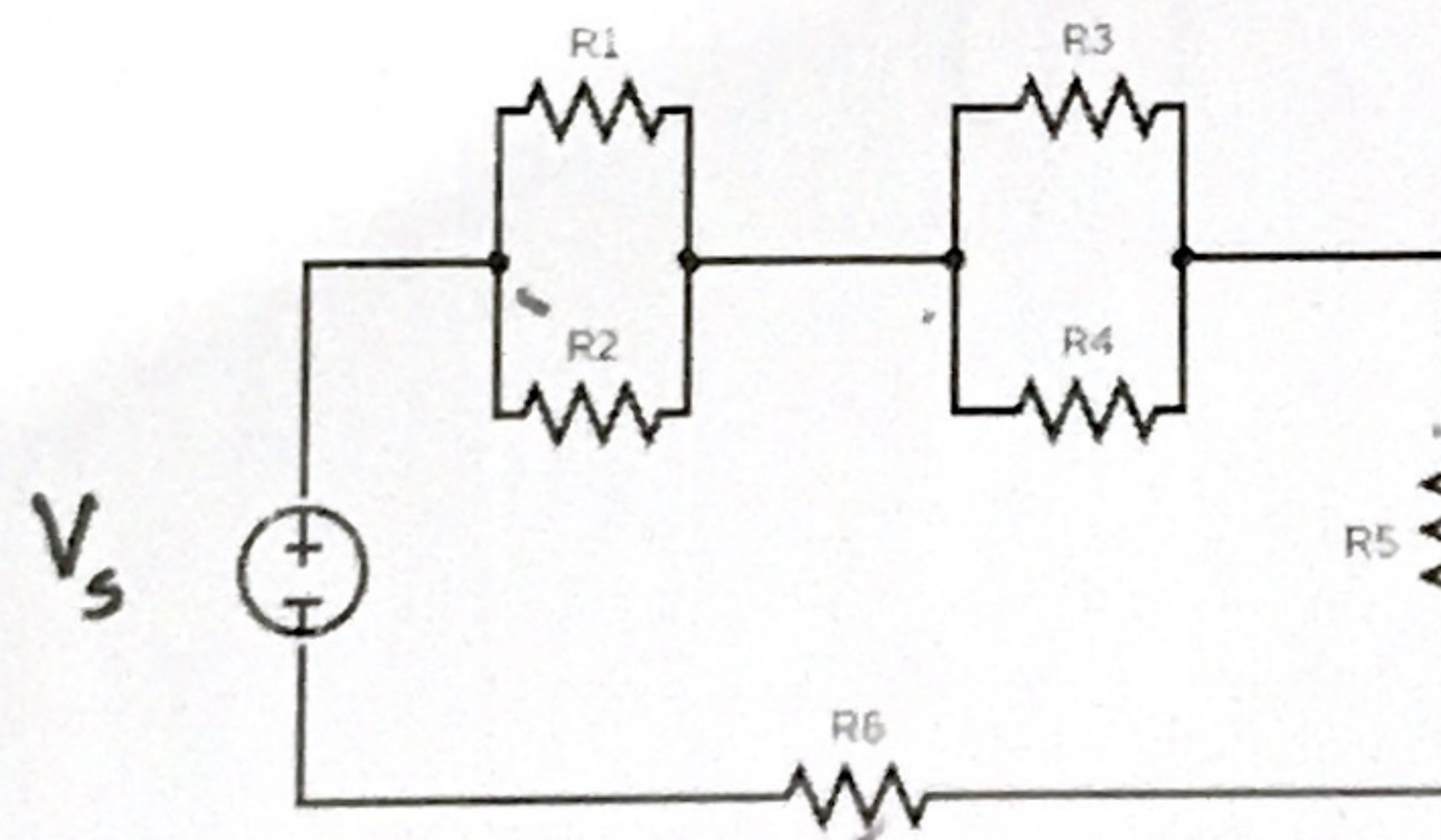
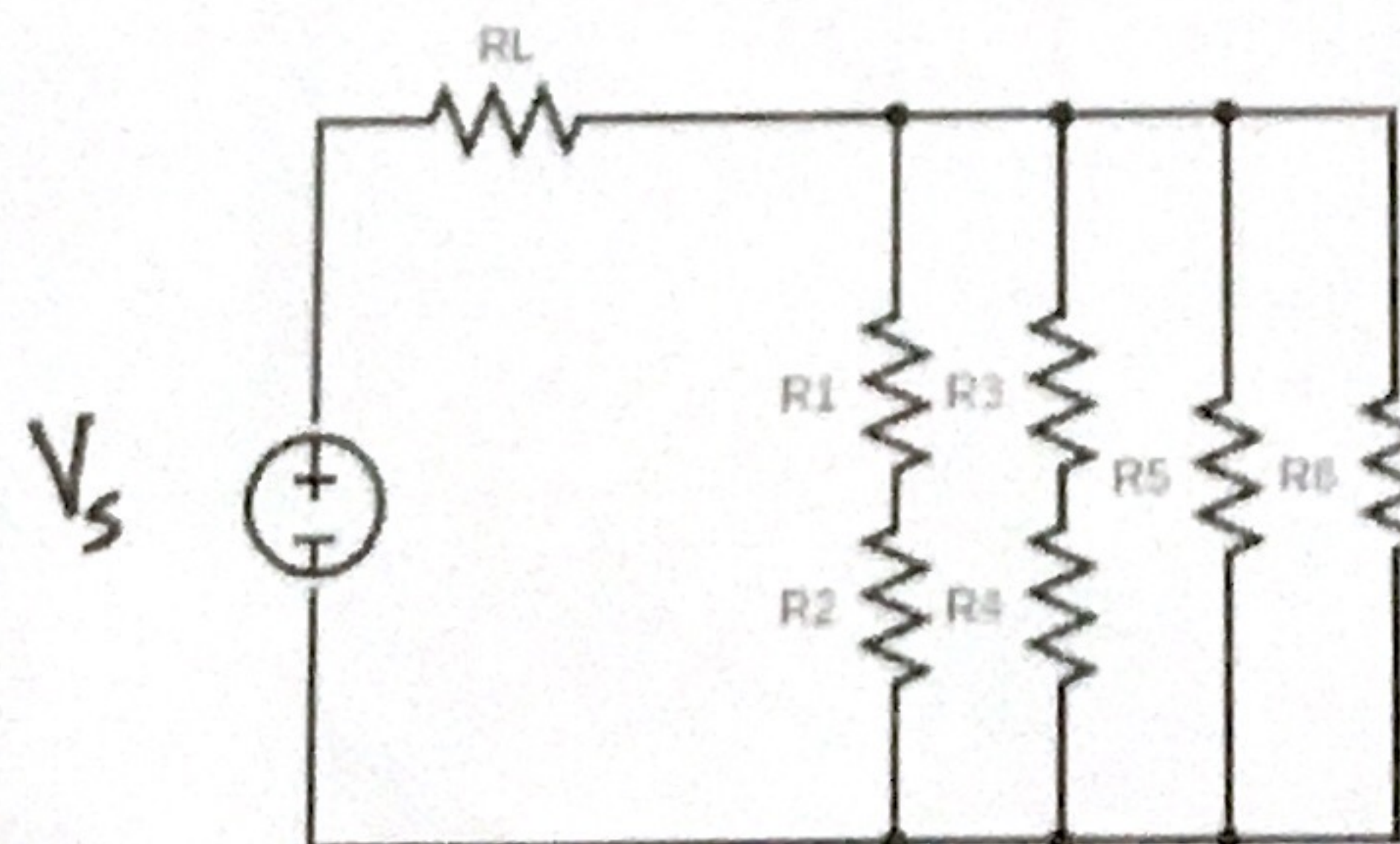


Fig.2. Circuit for Verification of the KVL

3. Verify Kirchhoff's Current Law for the following circuit. Iterate the process for two test cases. In the first test case, consider,  $V_s=12V$ , and in the second test case, consider,  $V_s=15V$ .



$2.7 \times 10^3$   
 $0.02$   
 $100$

Fig.3. Circuit for Verification of the KCL

4. Verify the Superposition Theorem for the following circuit. Consider the voltage from the first source,  $V_{s1}=10V$  and the voltage from the second source,  $V_{s2}=15V$ .

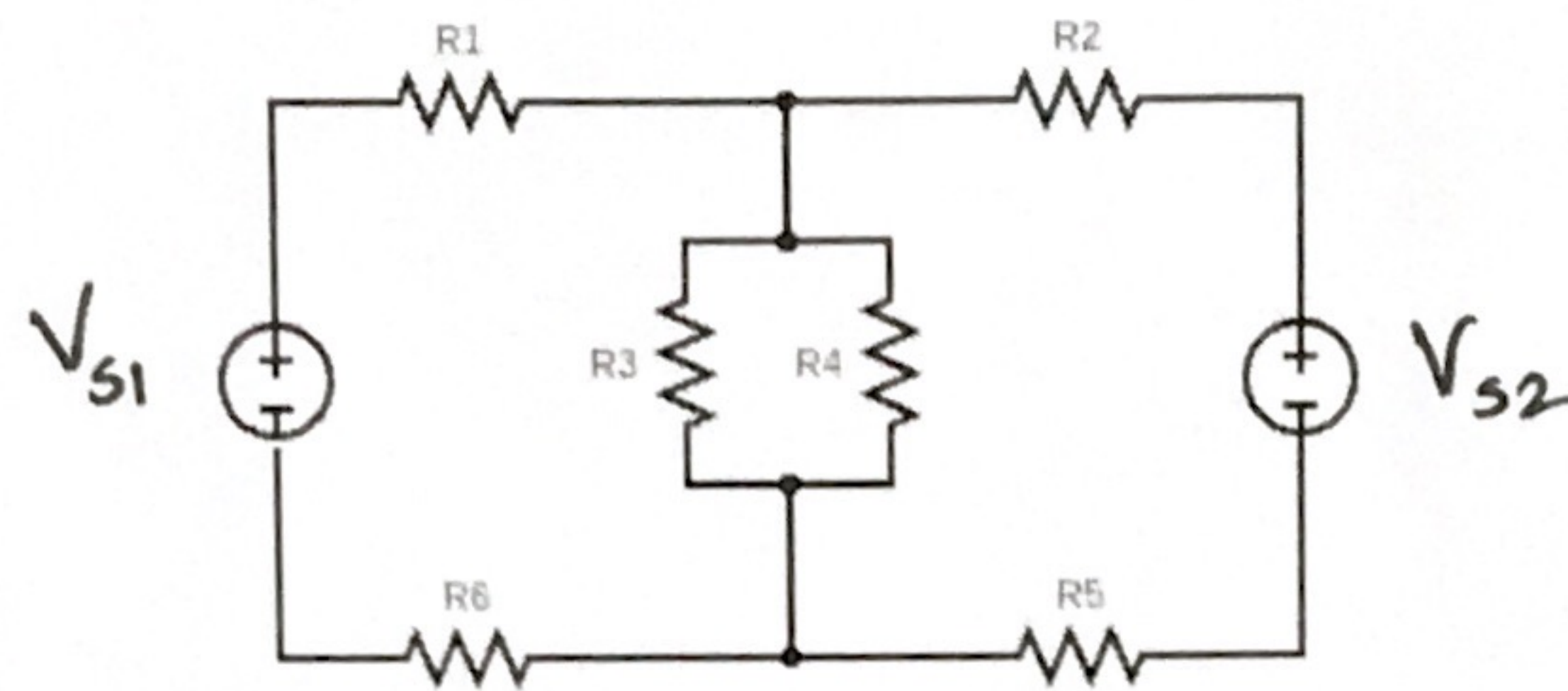


Fig. 4. Circuit for Verification of the Superposition Theorem

5. Verify the Thevenin's theorem for the following circuit. Consider the voltage of the source,  $V_s=15V$  in Fig. 5.

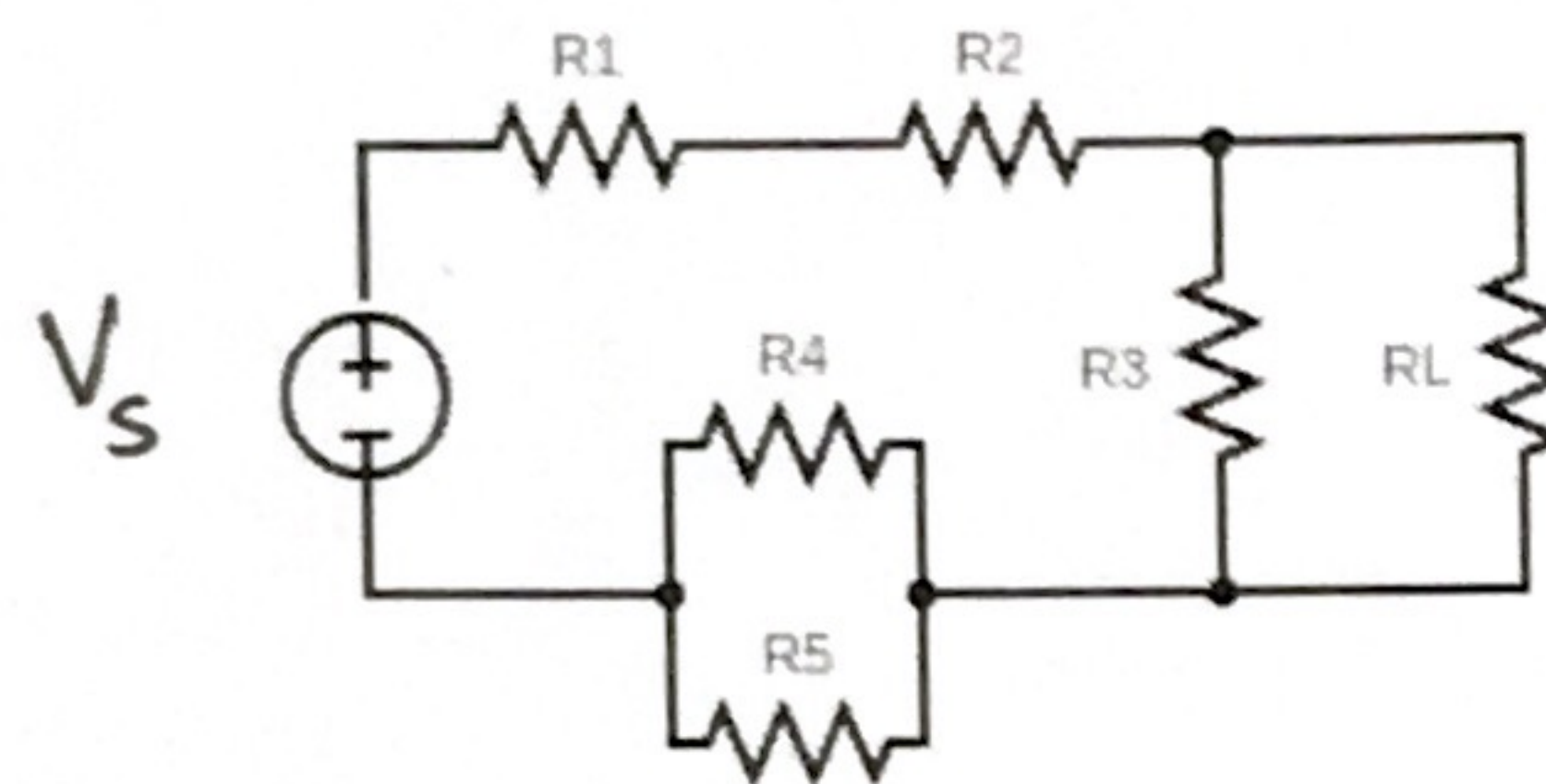


Fig. 5. Circuit for Verification of the Thevenin's Theorem

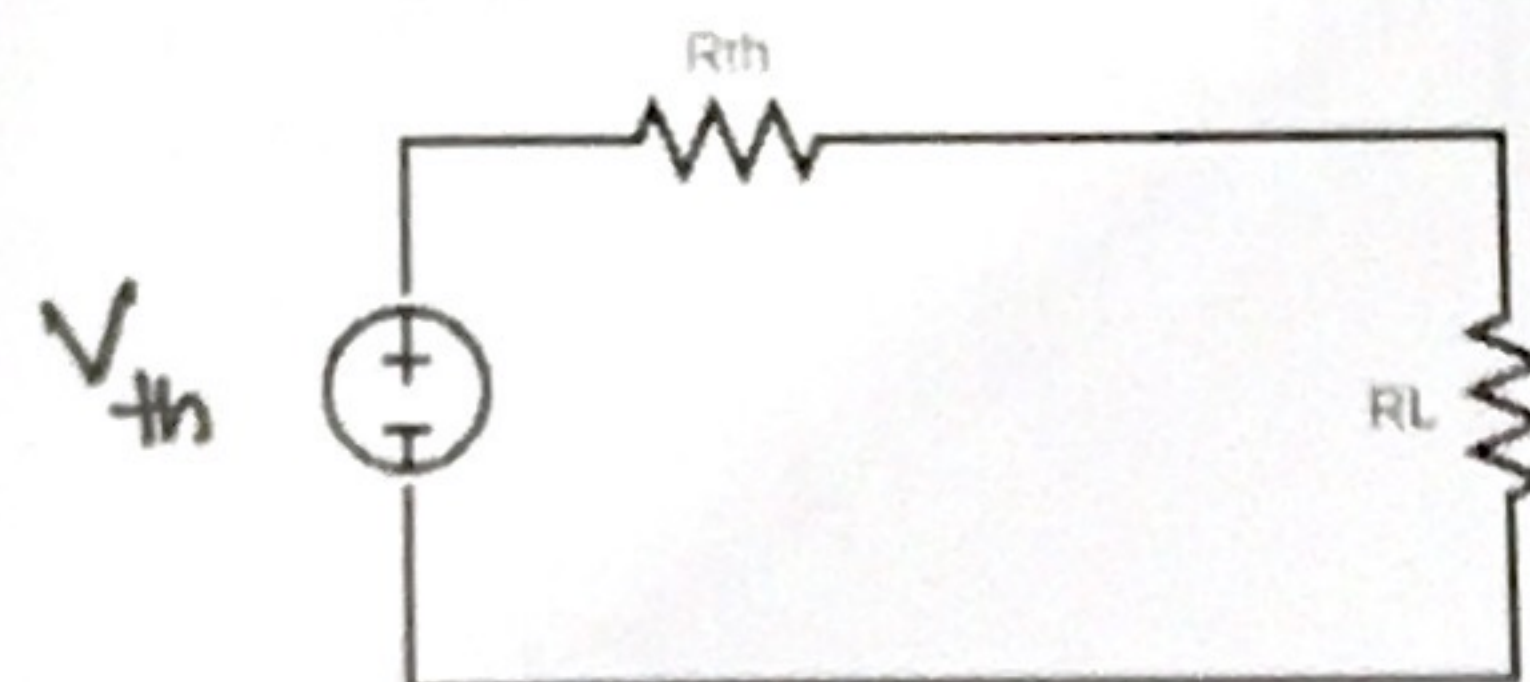


Fig. 6. Thevenin's Equivalent Circuit

6. Generate sinusoidal signals of 1KHz and 500Hz and measure the Amplitude, Root Mean Square (RMS) Voltage, Peak to Peak Voltage, Time Period and Frequency of the signals using an Oscilloscope. Calculate the Errors in Frequency and RMS Voltage measurement.
7. Perform an experiment for charging and discharging a capacitor and determine the Time Constant.